

EXPERIMENT 7

7 Study of Anomalous Behaviour of Water Using Hope's Apparatus

Aim

To study anomalous behaviour of water using Hope's apparatus.

Requirements

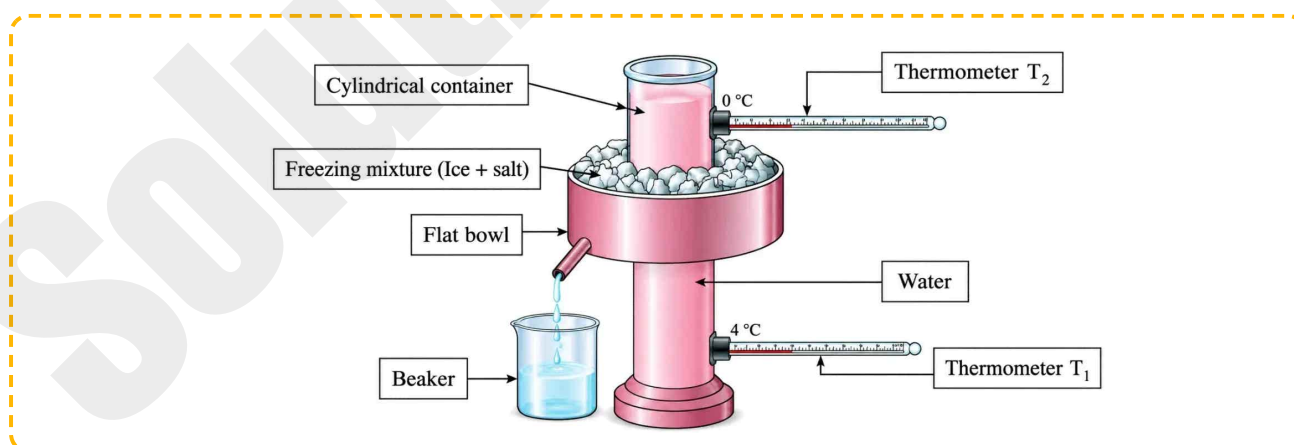
Hope's apparatus, ice, common salt, water, two thermometers, a beaker, and a stopwatch (or clock).

Procedure

1. Determine the least count of the thermometers.
2. Fill the cylindrical container of Hope's apparatus with cold water at about 10°C to 12°C .
3. Fill the flat bowl of the apparatus with a freezing mixture of ice and common salt.
4. Insert the two thermometers into the holes provided, so as to record the temperature of the water in the cylinder above and below the flat bowl respectively.
5. Note down the readings shown by the thermometers every minute and record them in the observation table.
6. Plot the graphs of temperature (T_1) versus time and temperature (T_2) versus time on the same graph paper.
7. Study the graphs that are obtained.

Diagram

(Label the various components in the given diagram - Hope's apparatus)



Observation Table

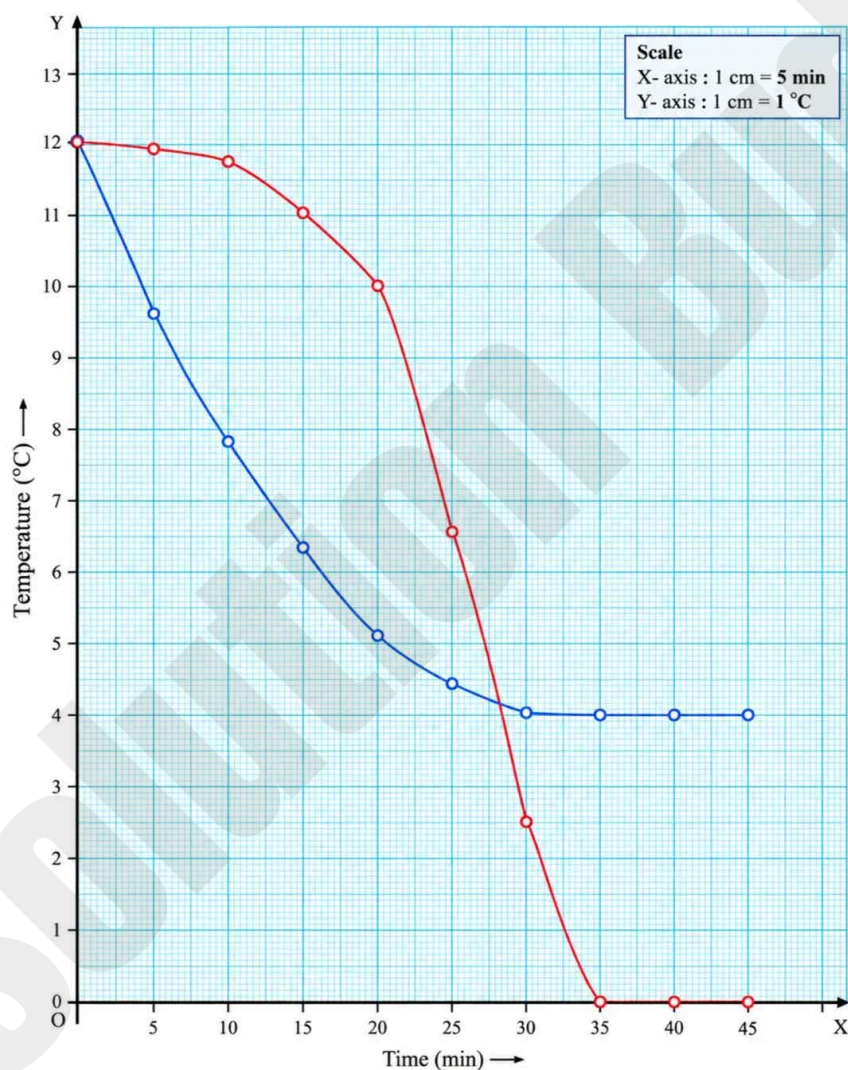
Least count of the thermometers = 1°C .

Time (min)	0	1	2	3	4	5	6	7	8	9	10	11	12	13
Temperature T_1 ($^{\circ}\text{C}$)	12					9.5					7.7			
Temperature T_2 ($^{\circ}\text{C}$)	12					11.9					11.7			

Time (min)	14	15	16	17	18	19	20	21	22	23	24	25	26	27
Temperature T_1 ($^{\circ}\text{C}$)		6.3					5.1					4.5		
Temperature T_2 ($^{\circ}\text{C}$)		11					10					6.5		

Time (min)	28	29	30	31	32	33	34	35	36	37	38	39	40	45
Temperature T_1 ($^{\circ}\text{C}$)			4					4					4	4
Temperature T_2 ($^{\circ}\text{C}$)			2.5					0					0	0

Graph of Temperature versus Time



Scale — X-axis: 1 cm = 5 min | Y-axis: 1 cm = 1 $^{\circ}\text{C}$

Inferences / Conclusion with Explanation

1. Initially, both the thermometers show the same temperature. In a short time, the temperature shown by the lower thermometer starts decreasing, while the temperature shown by the upper thermometer does not change very much.
2. This process continues till the temperature shown by the lower thermometer falls to 4°C and remains constant thereafter. This shows that in the temperature range 12°C to 4°C , the density of the water in the central part of the container goes on increasing and hence the water sinks to the bottom. It means that water contracts, i.e., its volume decreases as its temperature falls down from 12°C to 4°C .
3. As the temperature of the water in the central part of the container becomes less than 4°C , the temperature shown by the upper thermometer begins to fall rapidly to 0°C . But the temperature shown by the lower thermometer remains constant (4°C). Later, the reading shown by the lower thermometer decreases to 0°C .
4. In the temperature range 4°C to 0°C , the water moves upward. This shows that the density of water goes on decreasing in this range. It means that water expands, i.e. its volume increases as its temperature falls from 4°C to 0°C . The behaviour of water in the temperature range from 4°C to 0°C is called anomalous behaviour of water.
5. In the graph, the point of intersection of the two curves shows the temperature at which the density of water is maximum. This temperature is 4°C .

Precaution

Record all temperatures carefully.

Multiple Choice Questions

1. If the temperature of water is increased from 4°C to 10°C , then its

- (A) volume decreases and density increases
- (B) volume increases and density decreases
- (C) volume decreases and density decreases
- (D) volume increases and density increases

Answer: (B) volume increases and density decreases

2. At 4°C , the density of water is

- (A) 10 g/cm^3 (B) 4 g/cm^3
- (C) $4 \times 10^3\text{ kg/m}^3$ (D) $1 \times 10^3\text{ kg/m}^3$

Answer: (D) $1 \times 10^3\text{ kg/m}^3$

3. The density of water is maximum at

- (A) 0°C (B) -4°C (C) 100°C (D) 4°C

Answer: (D) 4°C

4. is used to study the anomalous behaviour of water.

- (A) The calorimeter (B) Joule's apparatus
(C) Hope's apparatus (D) The thermos flask

Answer: (C) Hope's apparatus

5. In this experiment, the point of intersection of the two curves in the graph corresponds to the

- (A) melting point of ice
(B) freezing point of water
(C) temperature at which the density of water is minimum
(D) temperature at which the density of water is maximum

Answer: (D) temperature at which the density of water is maximum

Remember

The graph of temperature of water against time shows that initially, both temperatures T_1 and T_2 are identical. The point of intersection of the two curves shows the temperature of maximum density.

Date: _____

Teacher's Signature: _____